

TOPIC 3/60

What is an LLM?

The brain behind ChatGPT.

Essentially, autocorrect on steroids.

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THE ACRONYM

We hear it everywhere, but what does it actually mean?
Let's break it down.

L = Large

L = Language

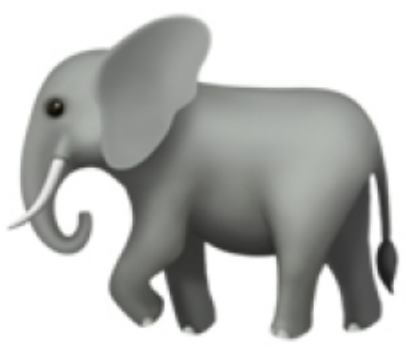
M = Model

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1. LARGE



Massive Scale

"Large" refers to two main things: the training data and the neural network itself.

The Data: Terabytes of text. Basically the entire public internet, books, and articles.

The Brain: Billions of "parameters" (synthetic synapses that store patterns).

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COMMUNICATION

2. Language

More than just English

These models are designed to understand, translate, and generate complex human languages.

Not just spoken words...

"Language" also includes Python, C++, Math, JSON, HTML, and even DNA sequences. Any data with strict sequential rules.

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THE ENGINE

3. Model

Mathematical Representation

It is NOT a database. It doesn't look up facts in a giant table or search the web directly.

It is a massive mathematical formula that computes the statistical probability of how words relate to each other.

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HOW IT THINKS

The Core Mechanic

Guess the next word

At its absolute core, an LLM is just playing a highly sophisticated game of autocomplete.

Example prompt:

"The quick brown fox jumps over the lazy ___"

> Calculates 99.9% probability for "dog".

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HOW THEY ARE BUILT

PHASE 1: PRE-TRAINING

Read the internet.

It learns grammar, facts, reasoning, and world knowledge.
(Costs \$100M+ and takes months).

PHASE 2: FINE-TUNING

Learn to be helpful.

Teaches the model to act like a polite Q&A assistant instead of just predicting text.

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FUN FACT

Tokens

LLMs don't actually read words; they read "Tokens". A token is roughly 3/4 of a word.

The word "Hamburger" might be split into [Ham] + [bur] + [ger]. This is exactly why AI image generators often struggle to spell words correctly!

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THE WEAKNESS

The Illusion of Thought

LLMs don't "think". If the math says a lie is statistically likely, they will output it as an absolute truth (Hallucination).

Memory Limits

They have a "Context Window". They can only remember a certain amount of text in a single conversation before they forget the beginning.

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